

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Manipulation of food species & agricultural energetics - Set 1

Mark scheme

AQA spec 3.5.1.2; Biological resources; textbook pp. 321-327

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding (2) • Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing (2) • Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) (2)	6
01.2	<i>1 mark for each point.</i> • AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies (1) • Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male (1) • Each embryo is implanted in a different surrogate female (1) • So the donor produces many offspring per year rather than one per pregnancy (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids (1) • Desired characteristics introduced without linked unwanted ones; faster (1)	2
02.2	<i>1 mark for each point.</i> • Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy (1) • Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage (1) • Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa (1)	3
02.3	<i>1 mark for each point.</i> • Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) (1) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria (1) • Detection in the human food chain from crops approved only for animal feed (1) • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use (1)	4
02.4	<i>1 mark.</i> • Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
03.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Extensive cereal 5.0; intensive cereal 3.0 (1) • Intensive beef 0.1 (1)	2
03.2	<i>1 mark for each point.</i> • Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) (1) • Each extra unit of input gives a smaller extra yield (law of diminishing returns) (1) • Beef has very low productivity and efficiency because energy is lost at an extra trophic level (1)	3
03.3	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> • Any input that increases productivity but requires the use of energy (usually fossil fuel) (up to 2) • Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery (up to 2)	3

Question	Marking guidance	Marks
03.4	<i>1 mark for each point.</i> • Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases (1) • Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	
04.1	<i>1 mark for each point.</i> • Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) (1) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted (1) • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion (1)	3
04.2	<i>1 mark for each point.</i> • Control of movement (housing / restricted grazing) so less energy is used in muscle activity (1) • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature (1) • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage (1)	3
04.3	<i>1 mark for each point.</i> • Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material (1) • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) (1) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot (1) • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	<i>1 mark.</i> • Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) (1)	1
05.2	<i>1 mark for each point.</i> • Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant (1) • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield (1)	2
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of valuable animals that die with genetically identical individuals (1) • Replacement of herds culled during a disease outbreak (1) • Production of large numbers of individuals with desirable characteristics (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
06.1	<i>1 mark for the correct answer.</i> • $1000 / 100\,000 \times 100 = 1\%$ (1)	1
06.2	<i>1 mark for each point.</i> • Some light is reflected by the leaves (1) • Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed (1) • Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) (1)	3
06.3	<i>1 mark for each point.</i> • Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) (1) • After four levels too little energy remains to support a further population (1)	2
06.4	<i>1 mark for each point.</i> • Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) (1) • Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat (1)	2
06.5	<i>1 mark for each point.</i> • Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) (1) • So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
07.1	<i>1 mark for each point.</i> • Yield increases as inputs increase (1) • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) (1)	2
07.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Second unit: $70 - 50 = 20$ units (1) • Fifth unit: $94 - 90 = 4$ units (1)	2
07.3	<i>1 mark for each point.</i> • The first units of input on each area give the largest yield increases (1) • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve (1)	2
07.4	<i>1 mark for each point.</i> • Some areas have much greater potential productivity so investment there is sensible (1) • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) (1)	2
07.5	<i>1 mark for each point.</i> • Yield per unit area is very high because of large inputs (1) • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	